CIRC HQ STUDENT HANDOUT | GRADE 5

Black Box Systems

Goal: Use controlled input-output tests to build and defend a model of a hidden system.

35 minutes | 8 steps | Computer Science and Programming | Playbook B meeting 17, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Present the model, one supporting result, and one remaining uncertainty.

You will use

- Teacher-prepared sealed systems
- Input pieces
- Measuring tools
- Data table
- Modeling paper

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the sealed system on its mat.
- Open the data table.

2 Step 2: Black box safety — 3 min

- Keep the box sealed.
- Use only the approved test holes.
- Move the box gently.

3 Step 3: Choose test actions — 4 min

- List three safe test actions.
- Change only one test action at a time.

4 Step 4: Run controlled tests — 8 min

- Try the first test action.
- Measure the result.
- Record the matching action and result.
- Repeat each action once.

5 Step 5: Draw the system — 7 min

- Draw one possible inside system.
- Connect each test action to a result.
- Label the hidden parts.

6 Step 6: Challenge the model — 5 min

- Choose one test that could show the drawing is wrong.
- Run that test.
- Revise the drawing if needed.

7 Step 7: Reset the box — 2 min

- Remove every test piece.
- Return the sealed system.

8 Step 8: Exit check — 3 min

- Write one claim about the hidden system.
- Cite one test result.

What to turn in

Your model, one supporting result and one question you still have.

No computer? Another way to do it

This lesson is already device free.

Finished early? Next challenge

Competing Model: Draw a different internal system that could produce the same evidence, then design a test that separates the two.

Stay safe

Do not pry, shake forcefully, or place any object into an opening not approved for testing.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.